

Report conversation

Given the following NRQL

```
SELECT count(newrelic.timeslice.value) as 'count',  
average(newrelic.timeslice.value) * 1000 AS 'duration'  
FROM Metric WHERE entity.guid =  
'MXxBUE18QVBQTELDQVRJT058NDE3Njg3NQ' and  
metricTimesliceName like '%hello_world%' FACET  
entity.guid, appName, metricTimesliceName  
SINCE 2 days AGO LIMIT MAX
```

With the following results

```
[  
  {  
    "metadata": {  
      "contents": {  
        "messages": [],  
        "contents": [  
          {  
            "function": "alias",  
            "alias": "count",  
            "contents": {  
              "function": "count",  
              "attribute": "newrelic.timeslice.value",  
              "simple": true  
            }  
          },  
          {  
            "function": "alias",  
            "alias": "duration",  
            "contents": {  
              "function": "binop",  
              "simple": true,  
              "binop": "*",  
              "left": {  
                "function": "average",  
                "attribute": "newrelic.timeslice.value",
```

```
"simple": true
},
"right": {
  "constant": 1000
}
}
}
],
},
"eventTypes": [
  "Metric"
],
"eventType": "Metric",
"openEnded": true,
"messages": [],
"beginTimeMillis": 1687188660000,
"endTimeMillis": 1687361460000,
"beginTime": "2023-06-19T15:31:00Z",
"endTime": "2023-06-21T15:31:00Z",
"guid": "02f43f90-9bfa-97f1-f4e2-68c8bb45f677",
"routerGuid": "02f43f90-9bfa-97f1-f4e2-
68c8bb45f677",
"rawSince": "1687188660000",
"rawUntil": "1687361460000",
"rawCompareWith": "",
"facet": [
  "entity.guid",
  "appName",
  "metricTimesliceName"
],
"offset": 0,
"limit": 2000,
"facetExpression": "tuple(entity.guid, appName,
metricTimesliceName)",
"timeAggregations": [
  "5 minutes"
],
"accounts": [
  1
]
```

```
},  
"facets": [  
  {  
    "name": [  
      "MXxBUE18QVBQTELDQVRJT058NDE3Njg3NQ",  
      "clm-demo-python",  
      "Apdex/Function/routes.app:hello_world"  
    ],  
    "results": [  
      {  
        "count": 4821  
      },  
      {  
        "result": 0  
      }  
    ],  
    "beginTimeSeconds": 0,  
    "endTimeSeconds": 0  
  },  
  {  
    "name": [  
      "MXxBUE18QVBQTELDQVRJT058NDE3Njg3NQ",  
      "clm-demo-python",  
      "Function/routes.app:hello_world"  
    ],  
    "results": [  
      {  
        "count": 4821  
      },  
      {  
        "result": 0.13733330529417476  
      }  
    ],  
    "beginTimeSeconds": 0,  
    "endTimeSeconds": 0  
  },  
  {  
    "name": [  
      "MXxBUE18QVBQTELDQVRJT058NDE3Njg3NQ",  
      "clm-demo-python",
```

```
"Function/routes.app:hello_world.<locals>.  
<lambda>"  
  ],  
  "results": [  
    {  
      "count": 4821  
    },  
    {  
      "result": 0.022502045411150227  
    }  
  ],  
  "beginTimeSeconds": 0,  
  "endTimeSeconds": 0  
},  
{  
  "name": [  
    "MXxBUE18QVBQTELDQVRJT058NDE3Njg3NQ",  
    "clm-demo-python",  
    "WebTransaction/Function/routes.app:hello_world"  
  ],  
  "results": [  
    {  
      "count": 4821  
    },  
    {  
      "result": 1.1917826209537603  
    }  
  ],  
  "beginTimeSeconds": 0,  
  "endTimeSeconds": 0  
},  
{  
  "name": [  
    "MXxBUE18QVBQTELDQVRJT058NDE3Njg3NQ",  
    "clm-demo-python",  
  
    "WebTransactionTotalTime/Function/routes.app:hello_world"  
  ],  
  "results": [  

```

```
{
  "count": 4821
},
{
  "result": 1.1917826209537603
}
],
"beginTimeSeconds": 0,
"endTimeSeconds": 0
},
],
"unknownGroup": {
  "results": [
    {
      "count": 0
    },
    {
      "result": null
    }
  ],
  "beginTimeSeconds": 0,
  "endTimeSeconds": 0
},
"totalResult": {
  "results": [
    {
      "count": 24105
    },
    {
      "result": 0.5086801185225691
    }
  ],
  "beginTimeSeconds": 0,
  "endTimeSeconds": 0
},
"performanceStats": {
  "inspectedCount": 14065,
  "responseTime": 124,
  "exceedsRetentionWindow": false
}
```

```
}  
]
```

What is the difference between the different measurements for `hello_world`? Why does `Function/routes.app:hello_world`, `WebTransaction/Function/routes.app:hello_world` and `WebTransactionTotalTime/Function/routes.app:hello_world` report different durations?

The different measurements for `hello_world` in the NRQL query results indicate different aspects of the performance metrics captured by New Relic. Let's break down each one:

1. `Function/routes.app:hello_world`

- This measurement represents the execution time of the specific function named `hello_world` within your application.
- In your provided results, it has an average duration of approximately 0.137 milliseconds.
- This focuses on the processing time of the code in that particular function and how long it takes to execute.

2. `WebTransaction/Function/routes.app:hello_world`

- This measurement represents the total time taken for a web transaction that involves the `hello_world` function.
- In your provided results, it has an average duration of approximately 1.192 milliseconds.
- It encompasses the time taken by the function itself (`Function/routes.app:hello_world`), plus any additional overhead or processing associated with the web transaction (e.g., middleware, networking).

3. `WebTransactionTotalTime/Function/routes.app:hello_world`

- This measurement is similar to `WebTransaction/Function/routes.app:hello_world`, but it might include even more elements. It represents the total time taken for a web transaction involving the `hello_world` function, including additional processing time.
- In your provided results, it also has an average duration of approximately 1.192 milliseconds.

- In this particular case, it seems like `WebTransaction` and `WebTransactionTotalTime` are reporting the same values, but in other cases, there might be differences based on more extensive processing, parallel executions, or background tasks linked to the transaction.
4. `Apdex/Function/routes.app:hello_world`
 - This measurement generally indicates the Apdex score, which is a measure of user satisfaction based on the response time of web applications and services. However, in this case, it seems to be showing the count without a score. Apdex is useful for gauging whether the performance is within acceptable levels for user experience.
 5. `Function/routes.app:hello_world.<locals>.<lambda>`
 - This measurement represents the execution time of a lambda function which is locally defined within the `hello_world` function.
 - It has a very small duration (approximately 0.023 milliseconds) as it's probably executing a very specific and limited piece of logic.

In summary, `Function/routes.app:hello_world` represents the time taken by the function itself, `WebTransaction/Function/routes.app:hello_world` represents the total time taken by the web transaction involving this function, and `WebTransactionTotalTime/Function/routes.app:hello_world` potentially includes even more elements associated with the transaction.